

### Questions by topic: Aquaculture - Set 1 (50 marks)

**Answer all questions in the spaces provided.**

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Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

**[3 marks]**

0 1 . 2

Describe how all-female rainbow trout are produced and explain why this is done.

**[3 marks]**

**0 1 . 3**

Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

**[4 marks]**

**Dissolved oxygen** .....

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**Water flow** .....

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**Day length** .....

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**Temperature** .....

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**10**

**0 2** Table 1 compares extensive oyster farming with intensive salmon farming.

**Table 1**

| Feature                | Extensive oyster farm                          | Intensive salmon farm                                     |
|------------------------|------------------------------------------------|-----------------------------------------------------------|
| Food input             | None - filter natural plankton                 | Fishmeal / plant pellets                                  |
| Stocking density       | Low, spaced on seabed                          | High, in sea cages / tanks                                |
| Breeding               | Wild spat settle on cleared seabed or hatchery | Selected adults stripped of eggs and milt                 |
| Disease / pest control | Removal of predatory crabs                     | Antibiotics, pesticides, wrasse, removal of diseased fish |
| Yield per hectare      | Low                                            | Very high                                                 |

**0 2 . 1** Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

**[4 marks]**

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0 2 . 2

Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 2 . 3

Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

10

0 3 . 1

Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

**Overfishing of wild fish for feed** .....

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**Sea lice and disease** .....

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**Escapes affecting wild gene pools** .....

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**Organic waste pollution** .....

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**03 . 2** Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

**[2 marks]**

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**10**

04 . 1

Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

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04 . 2

Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

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04 . 3

Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

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Do not write outside the box

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Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

**[9 marks]**

Extra space

**05 . 2** State what is meant by a **fed** aquaculture species.

**[1 mark]**

**10**

**END OF QUESTIONS**